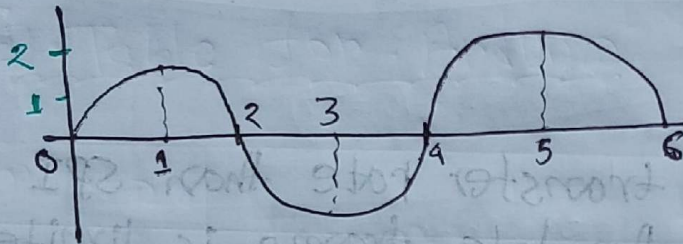


Analog to Digital & Digital to Analog conversion

□ signals can be two types

i) Analog

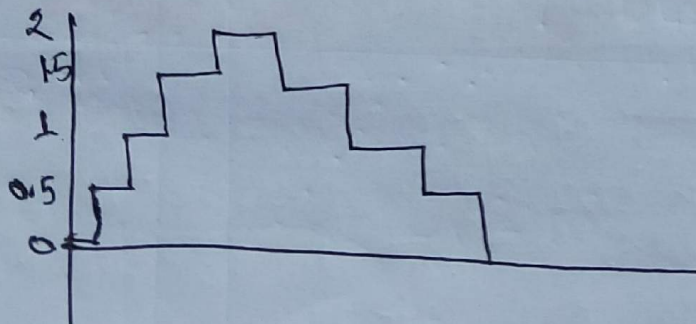
They are continuous



चित्र 1.1-1.2 वि
अर्थात् infinite value
होता है।

ii) Digital

The values are discrete here. ~~अतः~~ अतः infinite value को allow नहीं कर सकते, सिर्फ fixed value (integer and float both possible) को allow करते हैं।



□ Sensor can be two types:

1) Analog sensors:

Reading সূত্র শব্দ এও analog.

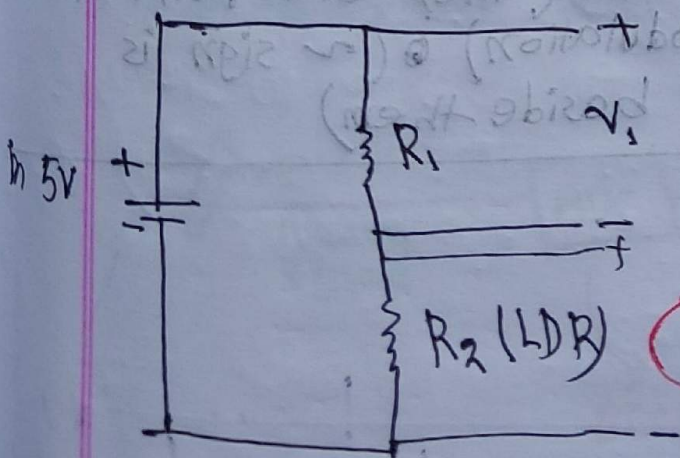
- i) light sensors
- ii) pressure sensors
- iii) temperature sensors.

LDR (Light dependent Resistor)

Used to find if the light is bright or dim
or what.

ਇਸ ਡਾਰਕ ਰਿਸ਼ਟਰ variable Resistance ਰਾਹੀਂ ਰਾਤ
ਡਾਰਕ ~~ਜਲਦੀ ਬਾਇ~~ ਜਲ ਜਦਲਾਸ light ਥਿ intensity
ਥਿ ਡਾਰਕ ।

light intensity $\uparrow$ LDR Resistance $\downarrow$ Voltage $\downarrow$



$$V_2 = V_{out} = V_{in} \left(\frac{R_2}{R_1 + R_2} \right)$$

ii) Digital sensors:

Gives digital output.

ਜਿਸ ਸਿਗਨਲ complex ਹੈ, ਵਾਸਤਵ ਵਿਸ਼ੇਸ਼ਤਾ data
ਜਿਸ ਨੂੰ analog, ਜਾਂ ਇਹ sensor ਨੂੰ ਸਿੱਧੇ
ਜਿਸਨੂੰ digital ਵਿੱਚ convert ਕਰਨਾ ਪੈਂਦਾ ਹੈ।

i) Ultrasonic sensor

ii) PIR sensor → Motion detector sensor

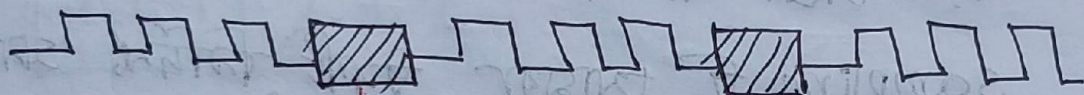
Table for Arduino

Operation	Pin
Digital Read/Write	D0-D13, A0-A5 (Analog pins can also work with digital value)
Analog Read	A0-A5 (They have ADC to convert to digital value)
Analog Write	D3, D5, D6, D9, D10, D11 (They are PWM → pulse with modulation) (~ sign is usually beside them)

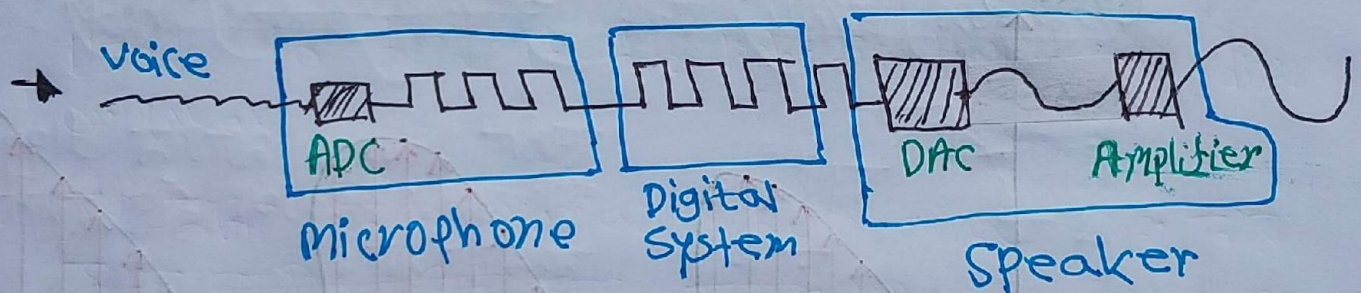
DAC use 5V

→ Analog signals get ~~diston~~ distorted/weaker as they travel distance.

→ Digital signals also does but ~~th~~ We have "repeaters" to amplify them.



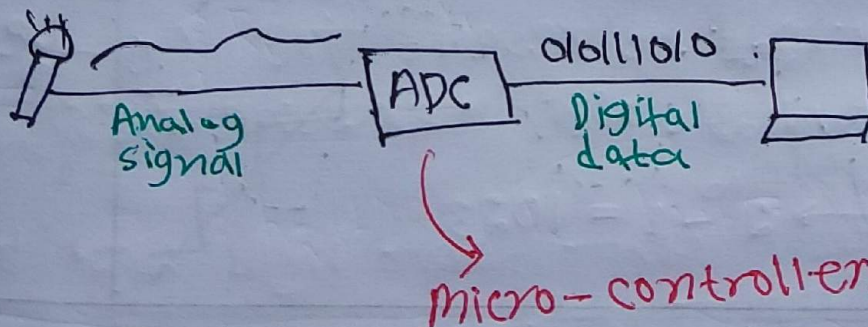
Repeaters



ADC

Analog to digital converter

pem → pulse code modulation use ~~शरीर~~ ADC



pcm work with 3 steps:

i) Sampling

તારફર એ analog signal-કે infinite વર્તમાન Value માટે, પરંતુ જો સમ Value નિર્ધારિત થાય તો શક્ય બને છે.

Sampling કે તારફર થઈ શકે તે માટે સમ analog signal ની value નિર્ધારિત થાય.

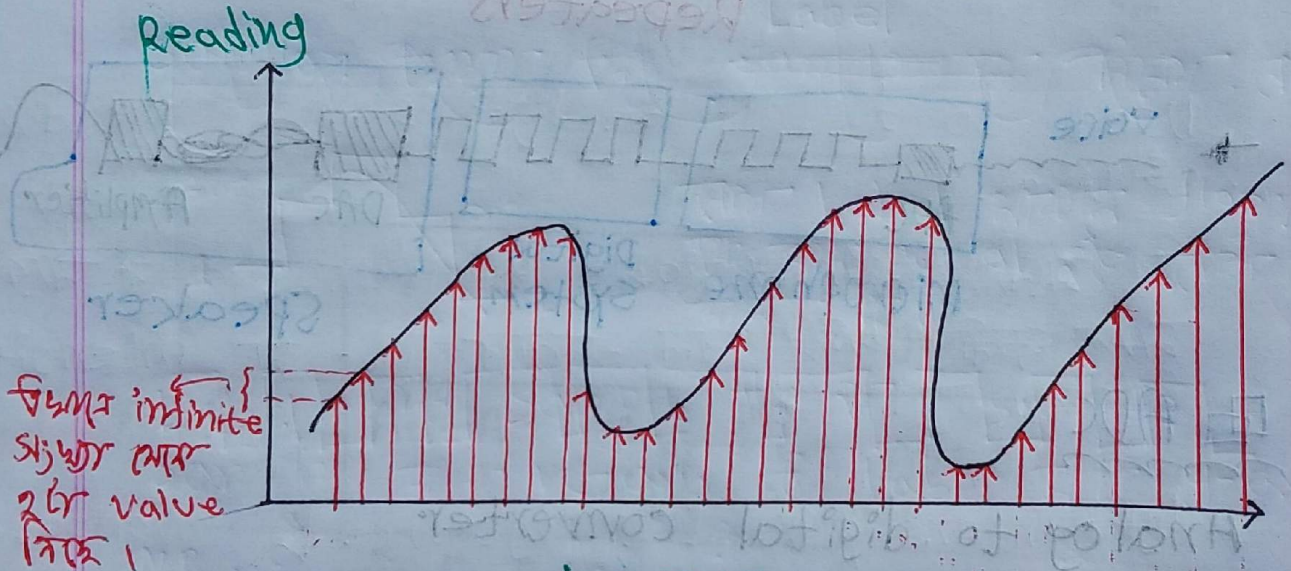


fig: Taking samples

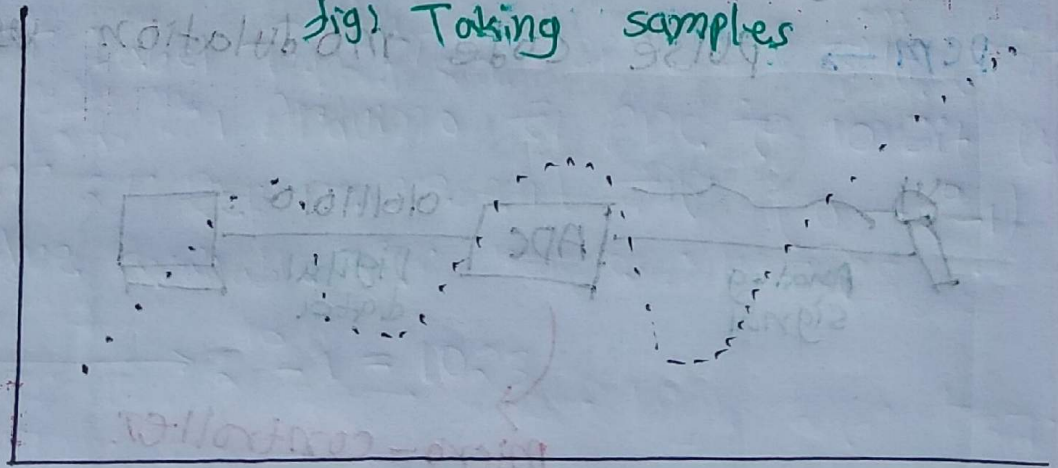
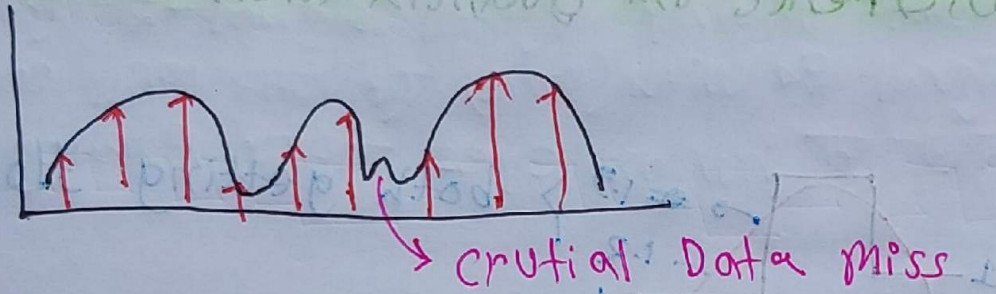


fig: output of sampling
(Each dot is a data point)

problem

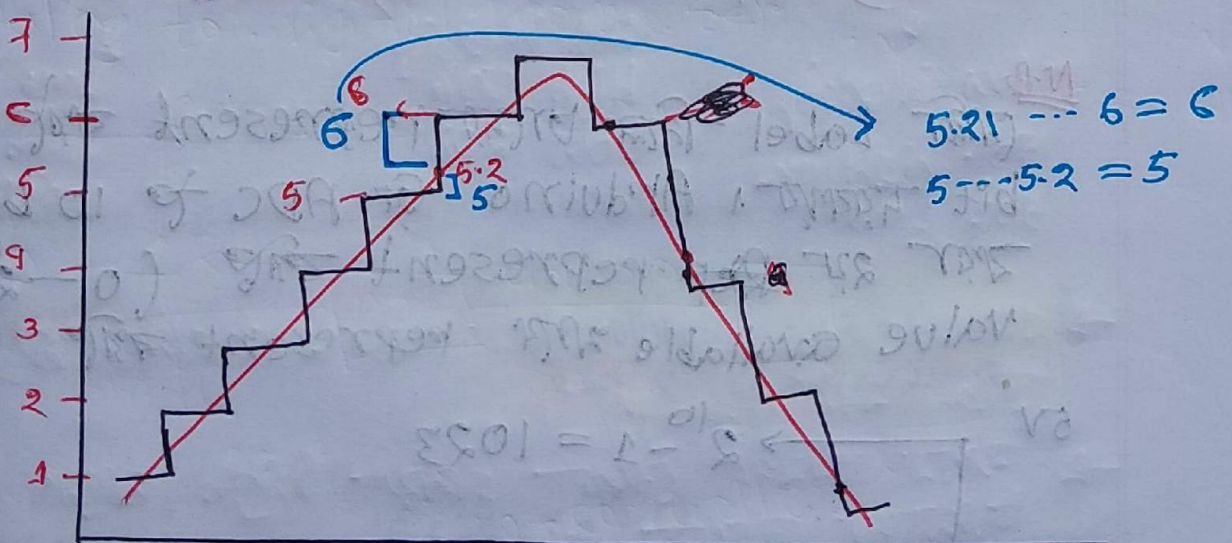
- i) यदि gap लीजें तो ~~error~~ sample में
जरूर possibility रहे कि crucial data miss होगा



- ii) Also if we take too many sample it
will ~~take~~ increase computational cost.

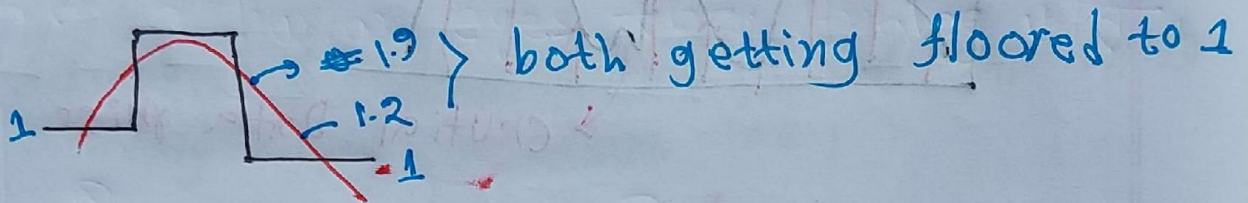
ii) Quantization

जिसके द्वारा Reading को discrete-तक लिए
जायगा • ceiling/flooring द्वारा

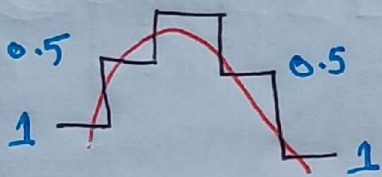


Problem

Flooring/ceiling करार करार actual data कि
small difference तक हो करे
Difference को quantization error.



Solution timer कि ~~sampling~~ निकार (कि)

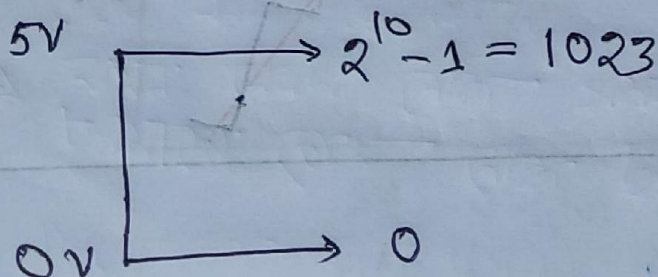


कि Label मानन floor/ceil
कि कर actual हो कि कर
करे कर।

Label कर sample कि कर sample
sampling कर x-axis कि, Label कर y-axis कि

N.B

कि Label निकार कर represent करे कि
bit कर कर। Arduino कि ADC को 10 bit we
कर कर ~~represent~~ represent करे $(0 - 2^{10} - 1)$ कर
value available कर represent करे



so, $3.7 \text{ V} = ?$

$5 \text{ V} \longrightarrow 1023$

$1 \text{ V} \longrightarrow \frac{1023}{5}$

$3.7 \longrightarrow \frac{1023 \times 37}{5}$

$= 757.02 \approx 757$ (always round value নিসা)

iii) Encoding

এখন আমরা step-২ এর পাওয়া value সুনিম্ন
Binary এ convert করব।

Ex: $3.7 \longrightarrow 757$ (1011110101) $\xrightarrow{757}$

$$\text{Analog Read (V) / ADC} = \frac{V_{in} \times (2^n - 1)}{V_{ref}}$$

V_{ref} = max voltage of the system.

summer 25 Q1(a)

Resolution $\Rightarrow (2^n - 1)$
12 bit Resolution $\Rightarrow n = 12$

$\text{Read A} = \text{Read B}$

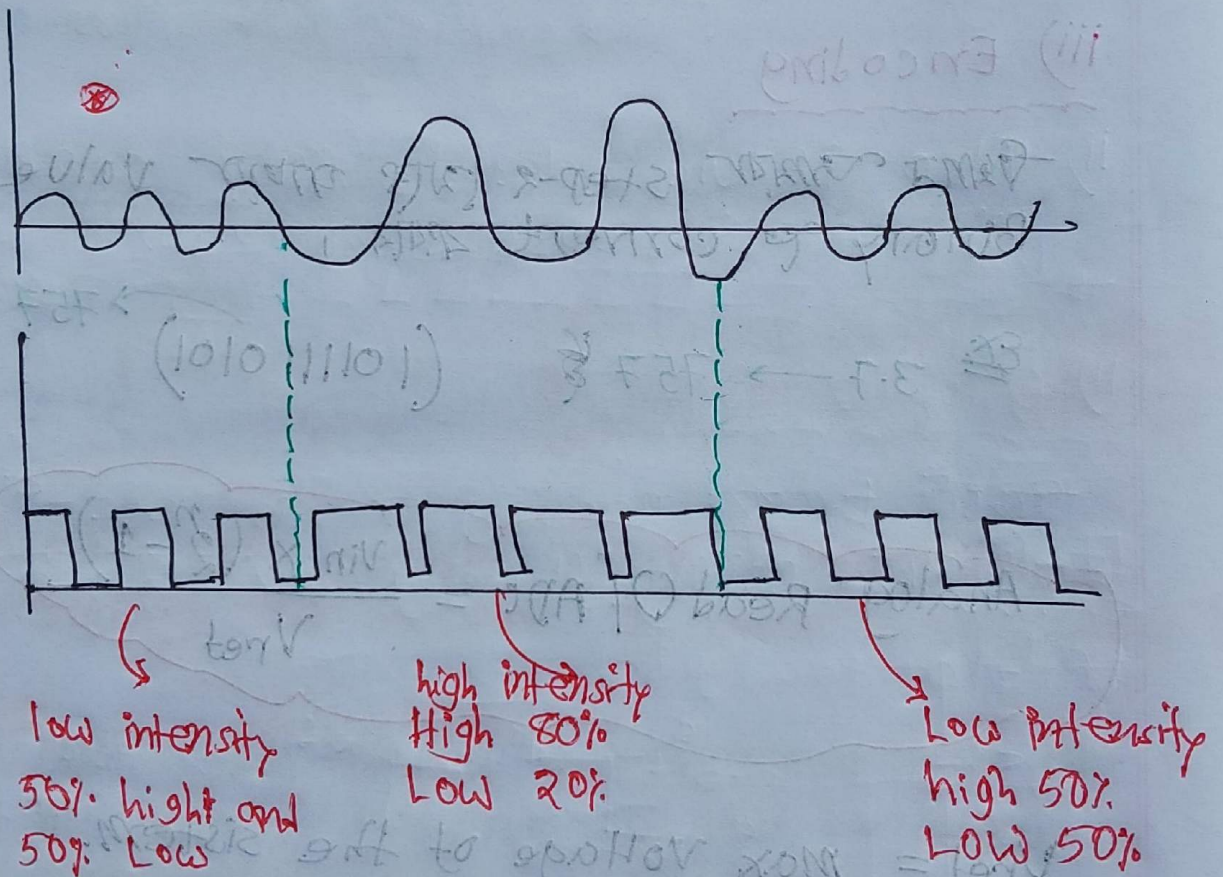
$\frac{(2^8 - 1) \times 8}{10} = \frac{(0.4 \times (2^{16} - 1))}{10}$

$\Rightarrow V = 128.5 \text{ V}$ (Voltage float হওয়ায়)

DAC

→ Digital to analog converter

বিক্রম মূলত digital signal-কে দিয়ে হাট চার signal দিয়ে with (time) এর ক্ষেত্রে analog control করে যে human eyes দিয়ে analog করে।

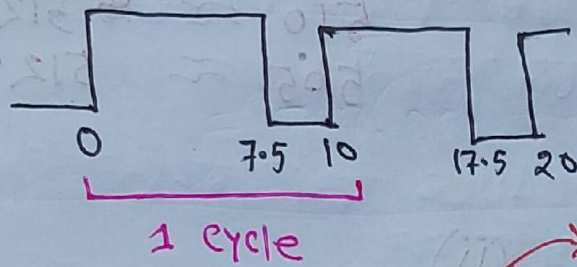


→ clock pulse দ্বারা পরিচালিত হওয়া হয় এবং সময়ের সাথে সাথে পরিবর্তিত হয়।

→ যখন পরিবর্তিত হয় তখন high থাকে তখন অন্ধকার (dark), যখন bright হয়ে ওঠে, DAC কাজ করে Using (pwm)

Duty cycle

→ एक cycle में signal high
होने का समय



$$DC(\text{duty cycle}) = \frac{7.5 - 0}{10} \times 100 = 75\%$$

high for stop
high for start.
cycle duration

$$DC = \frac{n}{2^n - 1} \times 100 = \frac{\text{high time}}{\text{Cycle time}} \times 100$$

Fall 25 Q1 (a)

$$n = 9$$

$$DC_T = DC_S$$

$$\Rightarrow \frac{488}{2^{12} - 1} \times 100 = \frac{122 \times 100}{2^n - 1}$$

$$\Rightarrow 2^n - 1 = \frac{122 \times 100 \times (2^{12} - 1)}{488 \times 100}$$

$$\Rightarrow 2^n - 1 = 1023.75$$

22th
class

$\Rightarrow 2^n = 1024.75$ (fraction value कि फल विह्वर
मकि आमाइ कि 2 वर power
आमाइ एकक ब्यादा)

$\Rightarrow 2^n = 1024$

$\Rightarrow n = 10$

Ex: $1022 \approx 1024$

$510 \approx 512$

$505 \approx 512$

(ii)

$$ADC = \frac{V_{in} \times (2^n - 1)}{V_{ref}}$$

$$= \frac{3.37 \times (2^8 - 1)}{5}$$

$= 171.87 \approx 172$ (आमाइ Round value कि)

$n \geq 0.5 \approx 1$
 $n \leq 0.4 \approx 0$

$$Dc = \frac{172 \times 100}{2^8 - 1}$$

$= 67.95\%$

Summer Q2 (a)

$$\frac{V_{in A} \times (2^n - 1)}{V_{ref A}} = \frac{V_{in B} \times (2^n - 1)}{V_{ref B}}$$

$$\Rightarrow \frac{8 \times (2^8 - 1)}{10} = \frac{0.4 \times (2^{16} - 1)}{V_{ref B}}$$

$$\Rightarrow V_{ref B} = \frac{10 \times 0.4 \times (2^{16} - 1)}{8 \times (2^8 - 1)}$$

$$= 128.5 \text{ V}$$

N.B

चूंकि mainly ADC of A चिह्न आलाह तो
 (य) वही Round करके exact Answer बताय
 जाय, नर बदलित फलन प्रदान नहि।

$$\Rightarrow \frac{8 \times (2^8 - 1)}{10} = \frac{0.4 \times (2^{16} - 1)}{V_{ref B}}$$

$$\Rightarrow 204 = \frac{26214}{V_{ref B}}$$

for fraction

आता Round

कराय 1

$$\Rightarrow V_{ref} = \frac{26214}{204}$$

$$= 128.5 \text{ V}$$